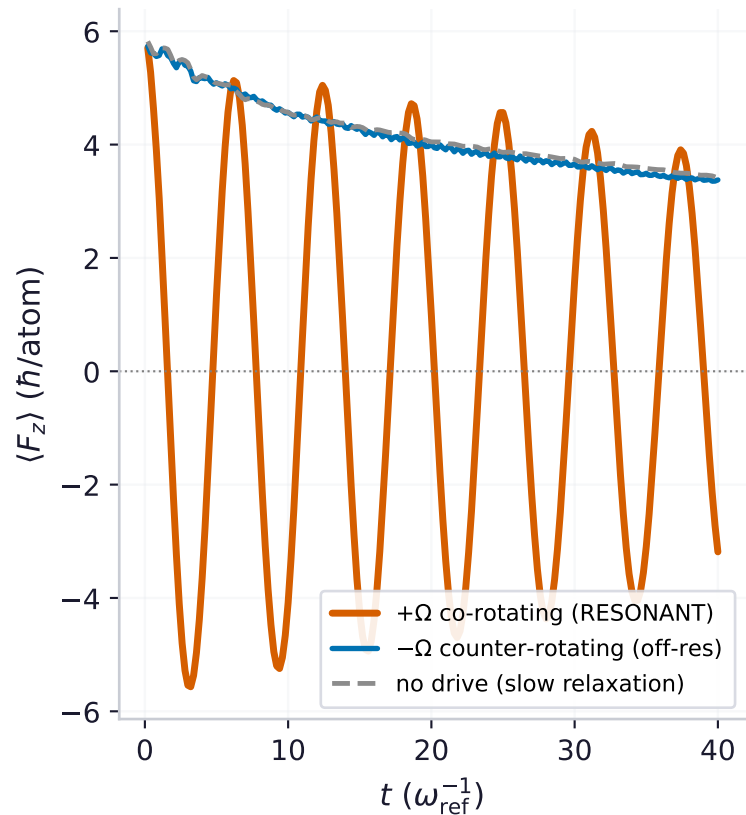
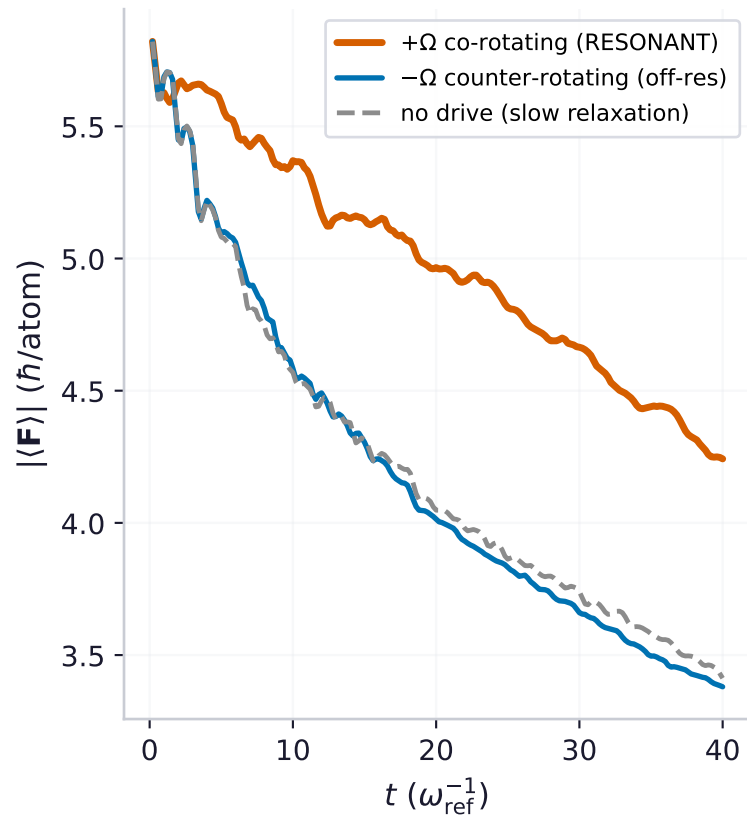


Field-UP one-sided excitation: metastable $m=+F$, large gap $\omega_L=5$ (relaxation slowed; only resonant $+\Omega$ drives) — ^{151}Eu $F=6$

(a) $\langle F_z \rangle$: only $+\Omega$ Rabi-flops



(b) $|\langle \mathbf{F} \rangle|$: resonant coherent, else depol.



(c) $\langle L_z \rangle$: vortices in the relaxation channel

